

Study of Seasonal Variations in the Quiet Time Geomagnetic Field at the Indian Antarctic Station 'Maitri'

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Abstract

The quiet-time daily variations of the geomagnetic field components X, Y, Z observed at the Indian Antarctic station Maitri (geog. 70°46'S, 11°45'E; geom. 66.84°S, 56.29°E), are discussed. The seasonal changes in these are examined in order to understand the atmospheric current systems which are responsible for such geomagnetic signatures. The signatures indicate that at quiet times AMI is under the influence of the Sq current loop in the southern hemisphere, and lies poleward of this loop. Deviations from this pattern are shown to be due to localised auroral activity adding its geomagnetic signature. The seasonal trend in the geomagnetic variations observed at Maitri suggest that the Sq current loop migrates northwards in the June solstice.

Introduction

It is now known that variations in the Earth's magnetic field are caused by electrical currents which flow at about 100 km altitude in the Earth's conducting upper atmosphere, mainly in the location known as E region of the ionosphere; a small portion of the currents flow in the higher ionospheric and magnetospheric regions. While these atmospheric currents contribute to about 2/3 of the observed magnetic variations, the remaining 1/3 is caused by electrical currents which are induced by the ionospheric currents in the conducting upper layers of the Earth's crust. The magnetic field variations in this paper are interpreted mainly in terms of the atmospheric current systems and not in terms of ground-induced currents.

At certain latitudes of the Earth, the configuration of the geomagnetic field is almost perpendicular to electric fields in the ionosphere, and this causes a great enhancement in the east-west directed ionospheric conductivity. This happens for example at equatorial latitudes within the $\pm 3^\circ$ belt, where the totally horizontal structure of the geomagnetic field results in the intense equatorial

electrojet currents; these are currents which flow in the E region in the east-west direction at about 100 km altitude due to enhanced Cowling conductivity. At auroral regions, between approximately 65° - 75° latitude in both hemispheres, the dawn-to-dusk electric field maps into the ionosphere to interact with an almost vertical magnetic field, and gives rise to strong east-west directed auroral electrojet currents; in this case they are Hall currents which also flow at E region altitudes of ~ 100 km. The auroral electrojet current during magnetically disturbed times can attain intensities of 10^4 amperes. This is partly due to the very high conductivity of the auroral ionosphere which is caused by the precipitation of charged particles (mainly energetic electrons) into the ionosphere from the distant magnetosphere. Such charged particle precipitation is much more on Disturbed days than on Quiet days, and is mainly confined to auroral and polar latitudes. The particle precipitation also gives rise to strong field aligned currents (FAC), markedly on Disturbed days at auroral latitudes; such FAC link the auroral ionosphere to the Earth's distant magnetosphere.

While charged particle precipitation plays a large role in ionospheric conductivity at auroral latitudes (markedly at night-time hours), there is also a certain contribution from solar ultraviolet (UV) radiation. This latter contribution is of greater importance during the Antarctic summer months of November, December, January and February, when the Sun hovers over the horizon for most of all of 24 hours. The charged particle contribution to ionospheric conductivity at Antarctica gains importance during the Antarctic winter months of May, June and July, when the Sun is by and large absent from the visible sky-limits. Fig. 1 is a graphical illustration of the variation in the number of hours of sunshine experienced by Maitri (MAI) over the different months of the year, and shows how the Antarctic winter months are totally devoid of solar UV radiation.

Data Acquisition

India operates a fluxgate magnetometer round-the year at Maitri with geographic coordinates $70^{\circ}46'S$, $11^{\circ}45'E$, and geomagnetic coordinates 66.84° , $56.29^{\circ}E$ (according to IGRF 1990). The fluxgate magnetometer records daily variations in the X, Y and Z vector components of the geomagnetic field, as well as pulsations in the same. The data are recorded in both analogue (on chart-recorders) and digital (on solid state data loggers) form. These variations have been studied during the year 1992 for magnetically Quiet days (as indicated by the Kp index of magnetic activity) of the various months, and the results are presented in this paper. An attempt is made to understand the atmospheric current systems responsible for these variations in terms of season,

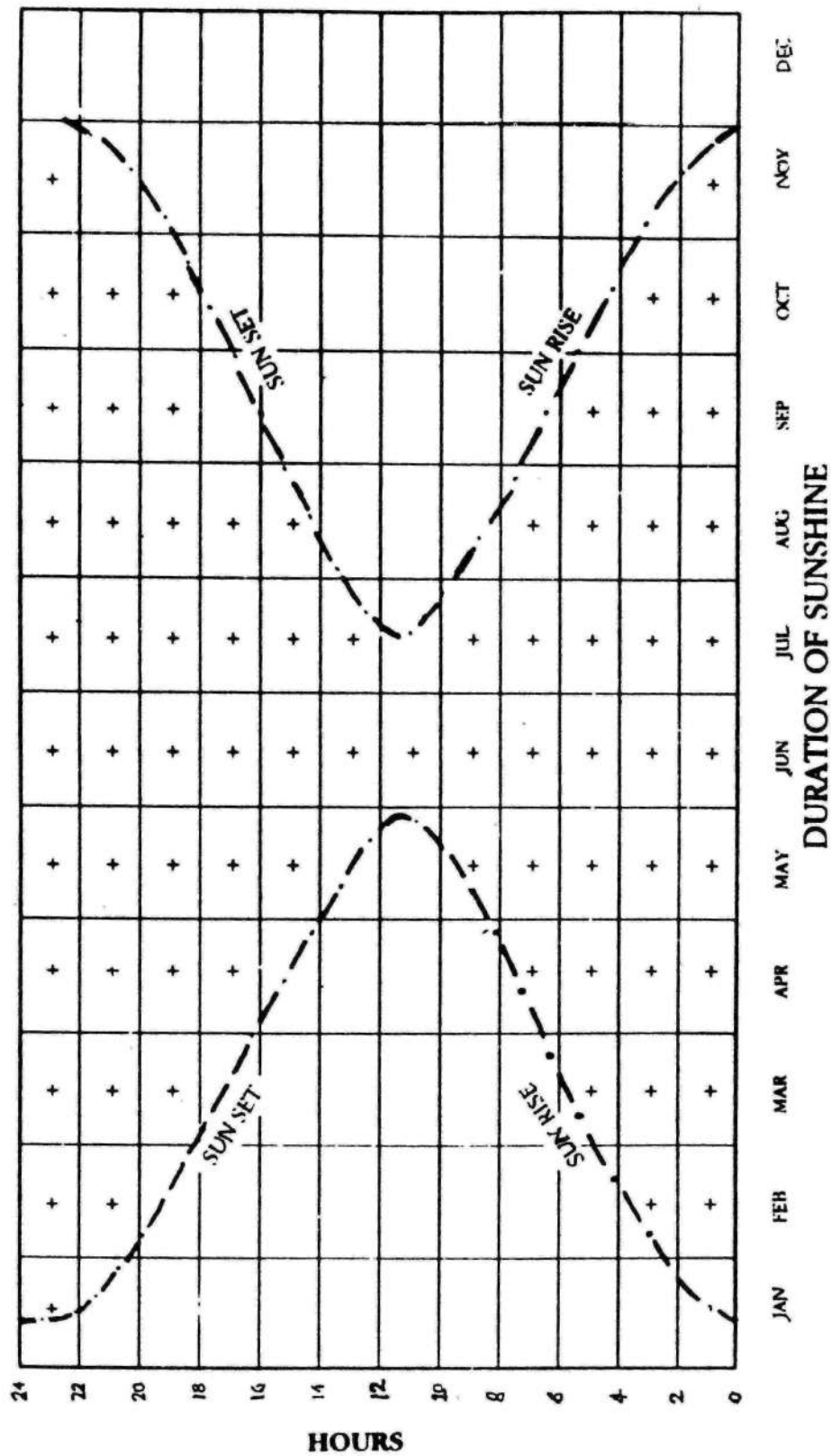


Fig. 1: The hours of sunlight and darkness over the Indian station Maitri (MAI) during various months of the year (from D.O.D. Annual Report 1988-89). Notice how MAI experiences almost total darkness during June mid to July, and total sunlight during December to January

and hence in terms of variations in the solar UV illumination received by the ionosphere above Maitri.

Observations and Interpretation

Fig. 1 portrays the hours of sunlight and darkness over the Indian Station Maitri during various months of the year. It is seen that during the period of about 1.5 months centered around mid-December, Maitri is continuously in sunlight, indicating that the ionosphere over Maitri is illuminated by solar UV all 24 hours of the day. In contrast, during the period of over 1.5 months centered around mid-June, Maitri is in total darkness, indicating that the ionosphere over Maitri never receives any solar UV radiation. This indicates "conditions like daytime" over Maitri around December and "conditions like night-time" around June. If the electrical conductivity caused by the ionization of the upper atmosphere by solar UV rays is to play a role in the ionospheric electric currents and hence in the magnetic signatures, we would expect maximum variation in the geomagnetic field during Antarctic summer and minimum variation during Antarctic winter at Maitri.

Seasonal Variation of the X (North-South) Component

Fig. 2 shows hourly values of the X component of the geomagnetic field measured at Maitri during Quiet days of the months Mar-Dec 1992. The diurnal variation of the X component on these Quiet days does show larger amplitude during Maitri summer than during Maitri winter. The magnetic activity level for the days selected is indicated by the Kp value indicated against that particular day. ΣKp is an index of magnetic activity at mid-latitudes and since ΣKp for the days considered has values between 5 and 10, the days selected are magnetically Quiet at mid latitudes. The horizontal lines for each month represent the Base Line Values (BLV) for the relevant component.

The seasonal variations of the X component shown in Fig. 2 indicate:

1. a typical Antarctic summer pattern from September through April 1992, with a minimum occurring around noon. The amplitude of this southward component is largest in the peak southern summer month of December, with a value of about 25 nT. The time of the noon minimum shifts from post-noon in October to pre-noon in December; this could be related to the time when the Sun attains minimum zenith angle in the relevant month. Some amount of positive (northward) variation is seen in the night hours, notably in the pre-midnight hours.

SEASONAL VARIATION OF X COMPONENT HOURLY VALUES MAITRI - 1992

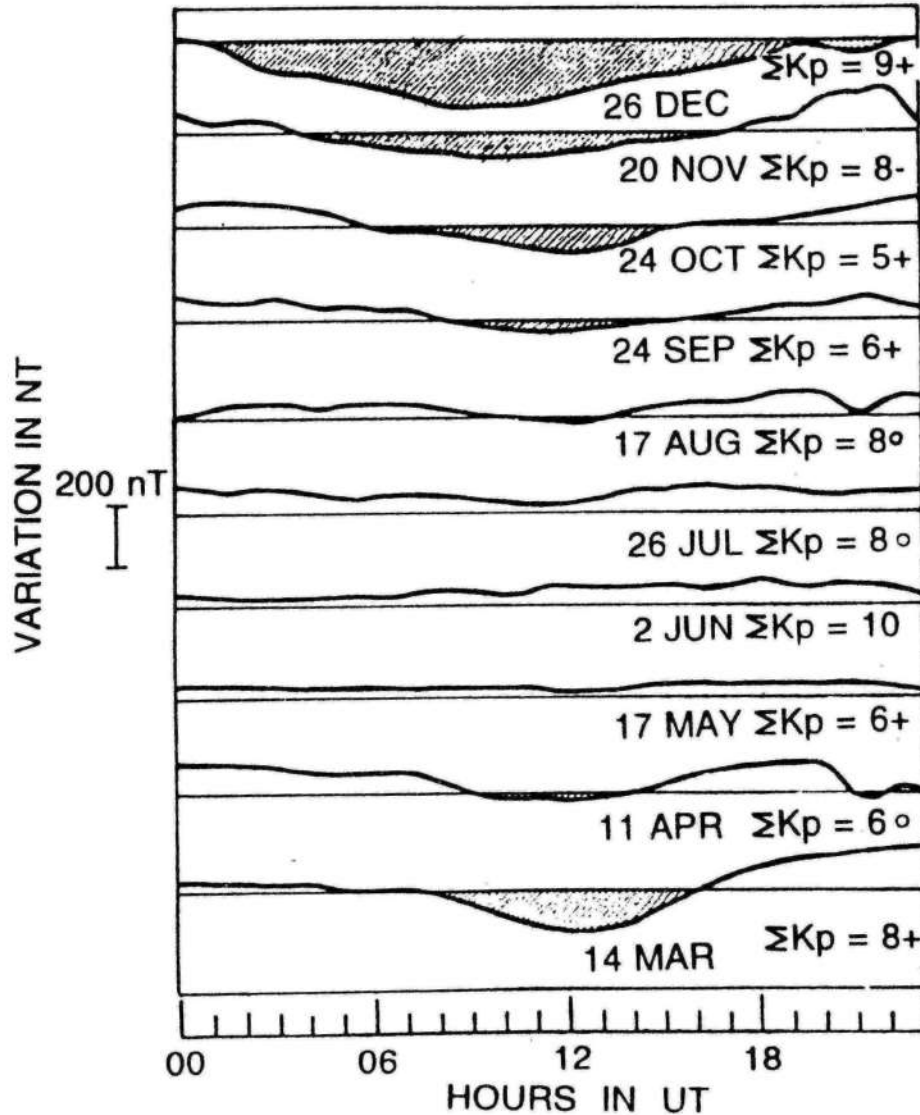


Fig. 2: Hourly values of the X-component (north-south) of the geomagnetic field measured at Maitri during Quiet days of the months March to December 1992

2. a typical Antarctic winter pattern from May through August 1992 when no negative (southward component) is seen and values remain positive over the major part of the day, notably between noon and midnight. This positive variation never exceeds 10 nT.

The summer pattern in the X component described above for these magnetically quiet days fits in well with Maitri experiencing the effect of the Sq (Solar quiet) current configuration at these southern latitudes. This is best seen

from Fig. 3. MAI occupies a sub-auroral position during quiet days, and hence it would not experience the effect of the Auroral Electrojet (AE) current system. What it experiences is the westward current limb of the Sq current loop, which leaves a southward geomagnetic signature; this is what is seen in Fig. 2. The Sq current loop is stronger in the Summer hemisphere, and is more intense in the dayside than in the nightside, hence the clear daytime minimum in X component in summer months.

The latitudinal position of the Sq current loops tend to follow the sun, and hence would move northwards in Antarctic winter months. The flat variation

CURRENT SYSTEMS WHICH INFLUENCE HIGH LATITUDES QUIET TIMES

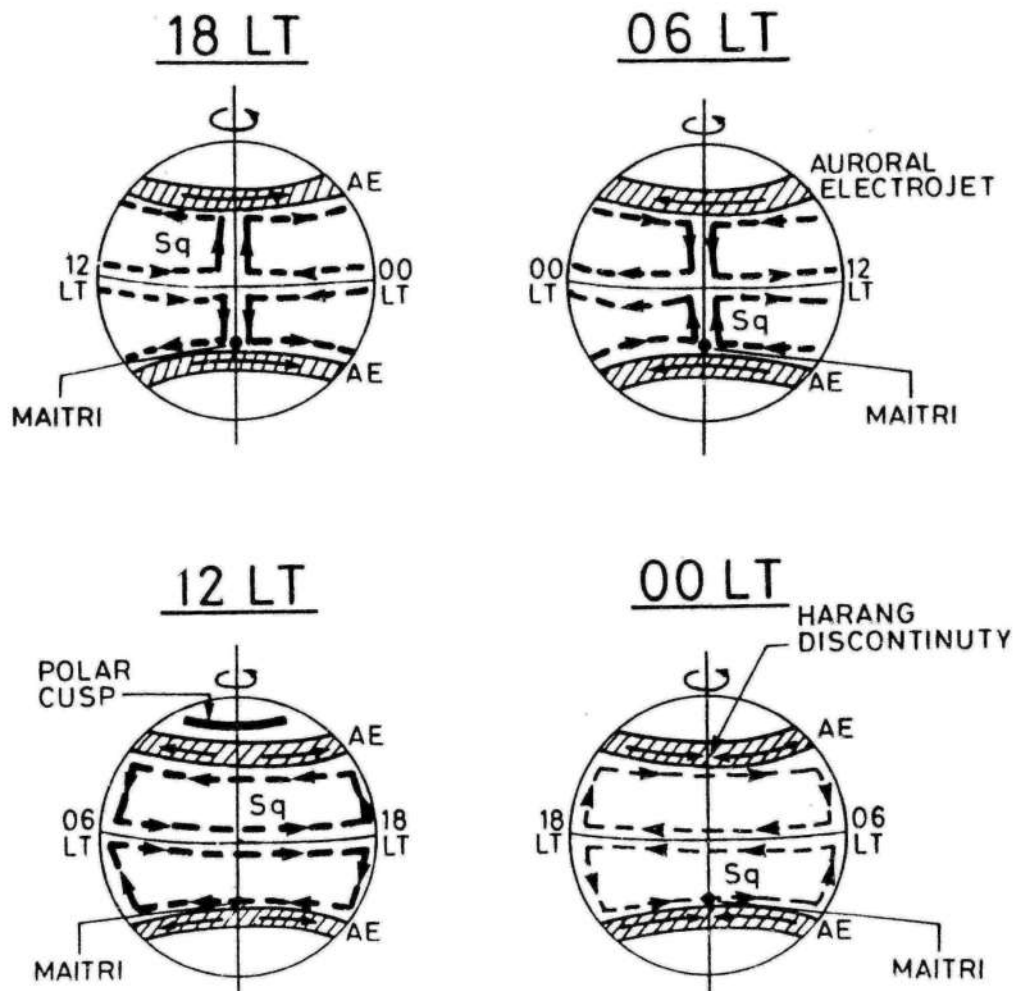


Fig. 3: Directions of Solar Quiet time (Sq) and Auroral Electrojet (AE) current systems during quiet days. The loop-shaped Sq currents are located in the ionosphere over low and mid-latitudes, while the AE flows in the auroral ionosphere

of the X component in these winter months is certainly not the signature of the Sq current. In addition MAI does not receive any direct solar UV radiation during the winter months. The electrical conductivity in the ionosphere over MAI would be caused by electrons and ions which are: (i) transported to these latitudes from higher latitudes which are illuminated by solar UV radiation or (ii) caused by the precipitation of energetic charged particles along geomagnetic field lines into the ionosphere from the magnetosphere.

Finally we notice that irrespective of the season there is a clear positive variation in the X component in the night hours. This is understandable in summer as the effect of the weak eastward Sq (cf Fig. 3) which MAI experiences in the night hours. In winter it can only be attributed to electrical conductivity caused by incoming charged particles. The night time positive is more enhanced on some days than on others; this feature will be discussed in the section on Discussion.

Seasonal Variation of the Y (East-West) Component

Variations in the Y component would arise from fluctuations in (i) the meridional north-south currents in the ionosphere or (ii) changes in the field-aligned currents (FAC) flowing into the ionosphere from the magnetosphere. Since the days chosen for this study are magnetically quiet days, the location of MAI would be sub-auroral and hence the contribution from the FAC may be very small. Hence, we would mainly seek causes in the meridional north-south component of the Sq current system (Fig. 3) which would influence MAI at quiet times.

Fig. 3 shows that the Sq current loops in the dayside southern hemisphere are directed such that MAI in the pre-noon hours should experience westward (negative) deflection, and in the post-noon hours an eastward (positive) deflection. This is exactly what is seen in Fig. 4 for the Antarctic summer months of September-December 1992, the same trend is seen for the equinoctial months of March and April, although all variations occur on the negative side of the Base Line Value (BLV). The maximum deflection on the negative and positive sides is about 25 nT. Thus the Y variations for the summer months are in keeping with those expected for a well-developed Sq current loop located above MAI. The Sq current loop in the night hours is a very weak one, and its geomagnetic signature on the ground may not be very discernible. Another feature seen in the Y variation in Fig. 4 (as in the case of the X variation in Fig. 2), is the progressive shift in the pre-noon negative from later hours in August to earlier hours in December. This could be connected with the Sun

attaining smaller zenith angle at earlier hours with the approach of Antarctic summer.

The Y variations during the Antarctic winter months of May, June, July and August follow a very different trend indeed; they are almost straight lines in the vicinity of the BLV (*i.e.* variation of less than 5 nT) suggesting that the Sq current loop has followed the Sun, and has moved away northward; the steady values of Y would suggest absence of influence of any variational atmospheric current systems on Quiet days.

SEASONAL VARIATION OF Y COMPONENT HOURLY VALUES MAITRI - 1992

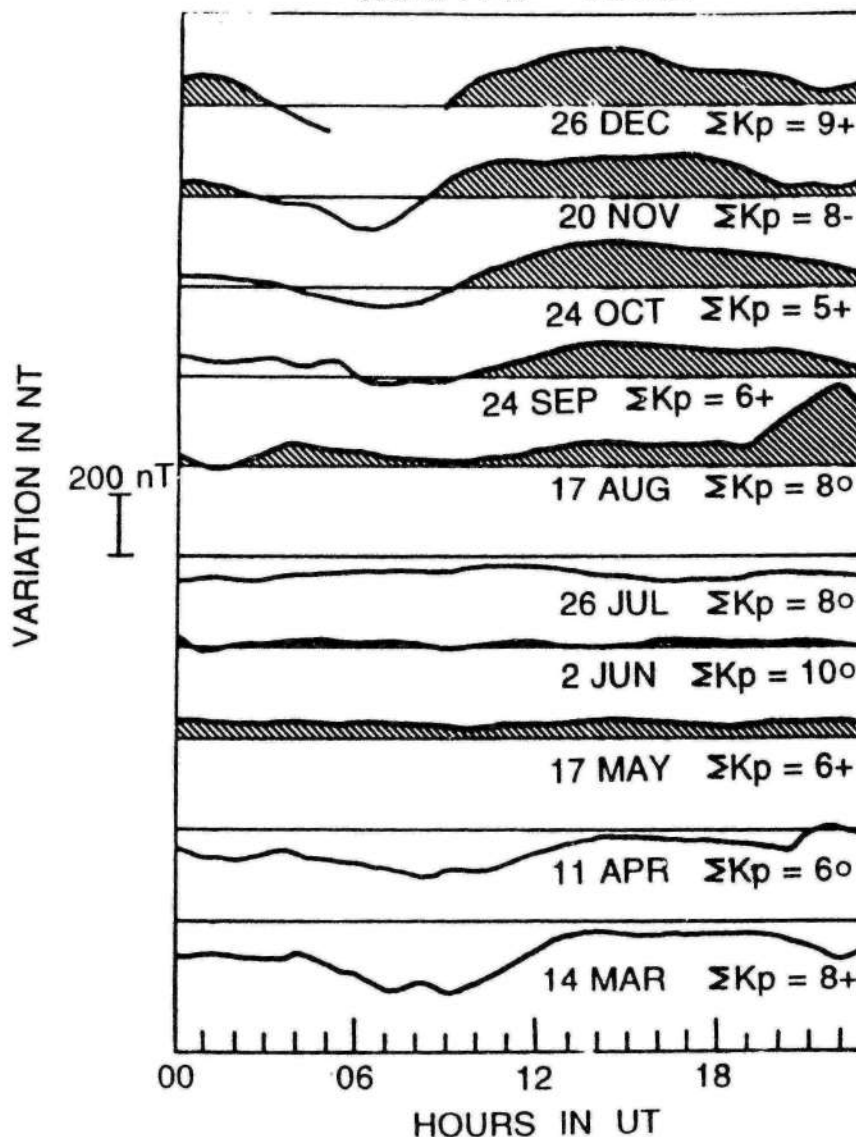


Fig. 4: Hourly values of the Y-component (east-west) of the geomagnetic field measured at Maitri during very quiet days of the months March-December 1992

The highly enhanced positive Y variation (>30 nT) just before midnight on 17 August cannot be connected with the Sq currents, and is discussed separately under the section on Discussion.

Seasonal Variation of the Z (Vertical) Component

Variations in the Z component can arise from an east-west current or a north-south current flowing the ionosphere. The type of Z variation, positive or negative, depends upon which side of the current the observation point is located. Hence Z geomagnetic variations are by-and-large difficult to interpret from single station data. A chain of magnetometers in the east-west and north-south directions is required for obtaining a latitudinal profile of the Z variations, and hence pinpointing the location of the current.

What can be seen from Fig. 5, is that the Antarctic summer months of Sep-Dec show a clear negative (*i.e.* upwards deflection) around noontime, attaining as much as 40 nT on 24 Oct. To some extent this trend is seen in the equinoctial months of March and April too. This negative variation could suggest that in these months MAI is located just outside (*i.e.* poleward of) the Sq current loop, and not inside it. The Z variations during the Antarctic winter months of May, June, July show a very different trend, with practically level values in the vicinity of the BLV. These straight line variations of Z as in the cases of the X and Y variations in Fig. 2 and Fig. 4 respectively, would show absence of influence from any variational current systems during Quiet days. As in the case of Fig. 4, the night-time deviation in Z around midnight hours on 17 Aug 1992 suggest a non-Sq source, and it is discussed in the following section.

Conclusion and Discussion

From the observations in Fig. 2, Fig. 4 and Fig. 5, it would suggest that :

- (1) There are marked seasonal changes in the geomagnetic variations at MAI, which clearly originate in a) the number of hours of solar UV which are experienced by MAI and b) the local time at which minimum zenith angle is attained over MAI.
- (2) The quiet time signatures of the X and Y components suggest that during Antarctic summer MAI experiences the geomagnetic signatures of the Sq current loop in the southern hemisphere. The Z signature on these Quiet days suggests that MAI lies just poleward of the Sq current loop, not within it.

SEASONAL VARIATION OF Z COMPONENT HOURLY VALUES MAITRI - 1992

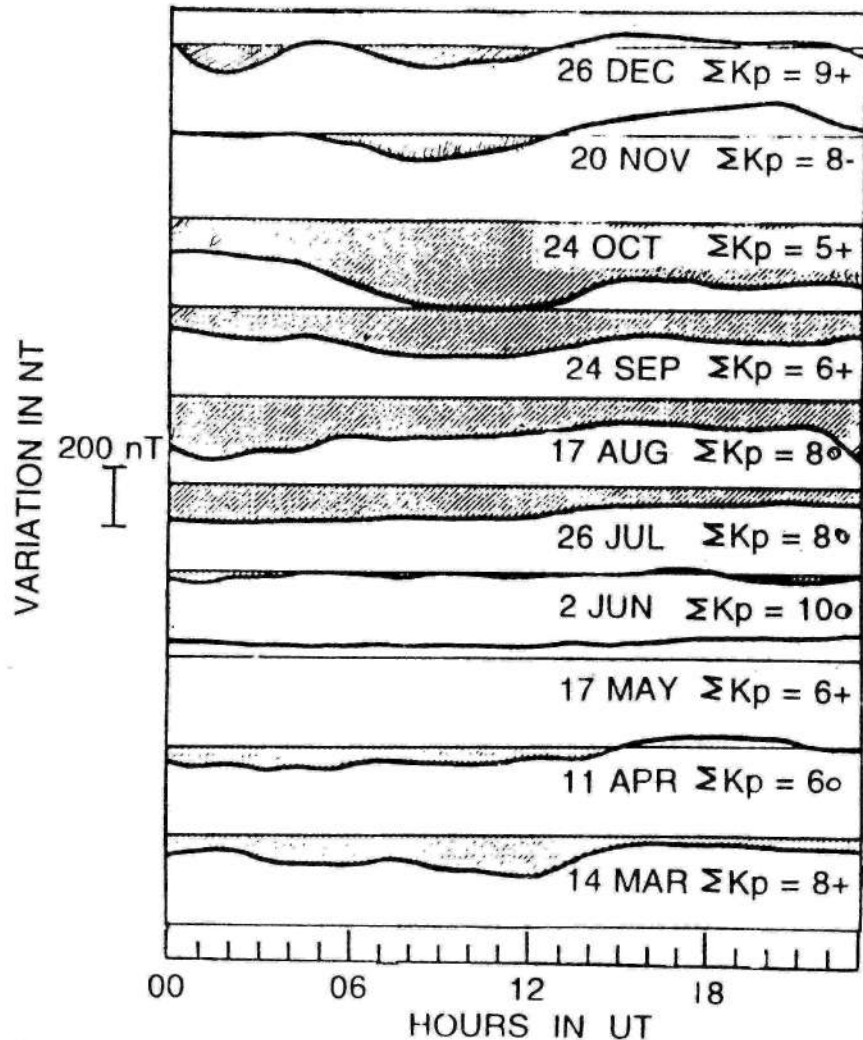


Fig. 5: Hourly values of the Z-component (vertical) of the geomagnetic field measured at Maitri during very Quiet of the months March-December 1992

- (3) The Antarctic winter months are characterised by absence of variation in the X, Y, and Z components suggesting that the Sq current loop in the southern hemisphere may have followed the sun and migrated slightly northward; subsequently the current system influencing the geomagnetic field at MAI on these Quiet days is not a variational one.
- (4) Some of the variations experienced by MAI in the night hours, *e.g.* ΔX on 20 Nov 1992, ΔY and ΔZ on 17 Aug 1992 suggest that there is a night time current source which is not characteristic of magnetically Quiet days. This aspect is discussed later on.

The change in the number of hours of solar ultraviolet radiation experienced by a location such as Maitri is substantial as can be seen from Fig. 6. The upper block illustrates the orbital path of planet Earth in its annual excursion around the Sun. The Summer (June) solstice represents Antarctic winter and is enlarged for convenience in the lower block. In this season the angle of 23.5° formed between the Earth's axis of rotation and the arc of day-night separation, is such as to leave Antarctica in total darkness over 24 hours of the day, while its counterpart Arctic region is in total daylight for 24 hours of the day. Just opposite would happen during the December solstice, leaving Antarctica in total sunlight and its counterpart Arctic region in total darkness. Thus the differing hours of solar illumination experienced by Maitri between summer and winter clearly influence the state of ionization, and the electric current which flow in the ionosphere over Maitri. The intensity of the E region ionospheric currents, and hence the intensity of the geomagnetic variations observed by Maitri, is a function of changing season.

Ionospheric Currents and Conductivities over Maitri

The interpretation of ground-based geomagnetic variations in terms of atmospheric current-sources dates back to Gauss (1838). These ideas were clarified by Balfour Stewart (1878), and expressed quantitatively by Schuster (1889). The first world maps of the Sq current loops in the ionosphere were charted from worldwide geomagnetic observations by Chapman and Bartels (1940), but these essentially took care of latitudes between $+60^\circ$. The ionospheric current systems above 60° i.e. essentially at auroral and polar latitudes were charted again from ground geomagnetic variations by Obayashi (1967). These current systems were classified as DP1, DP2 (disturbance polar), and DS (disturbance solar), and shown to be generated through the direct interaction of the solar wind with the geomagnetic field (Obayashi 1967). The interpretation of high-latitude and auroral geomagnetic variations in terms of ionospheric and field-aligned currents has since been carried out by many workers (Sato, 1965; Kisabeth and Rostoker 1973; Wiens and Rostoker 1975; Buchert et al. (1990); Andre and Baumjohann (1982); Aikio and Kaila (1996). These interpretations greatly enhance understanding of the energy coupling between the ionosphere and magnetosphere.

The currents which are responsible for the geomagnetic variations observed at MAI are clearly of two types 1) the ionospheric currents, namely Sq currents at Quiet times and the Auroral Electrojet (AE) currents at Disturbed times, and 2) the field aligned currents (FAC). These currents are dependent on the conductivity of the ionospheric region concerned, and on the electric field

in the region. The Sq currents depend on both Pedersen (σ_1) and Hall (σ_2) conductivities, while the AE currents are governed by the Hall conductivity (σ_2) which has a direction perpendicular to both electric (E) and magnetic (B) fields.

The FAC link the distant magnetosphere with the ionosphere and are governed by the longitudinal or direct conductivity (σ_0).

The height variations of these conductivities are discussed by Rishbeth and Garriott (1969). The Hall conductivity attains maximum value at all latitudes

DIFFERENTIAL SUMMER-WINTER SOLAR UV RADIATION

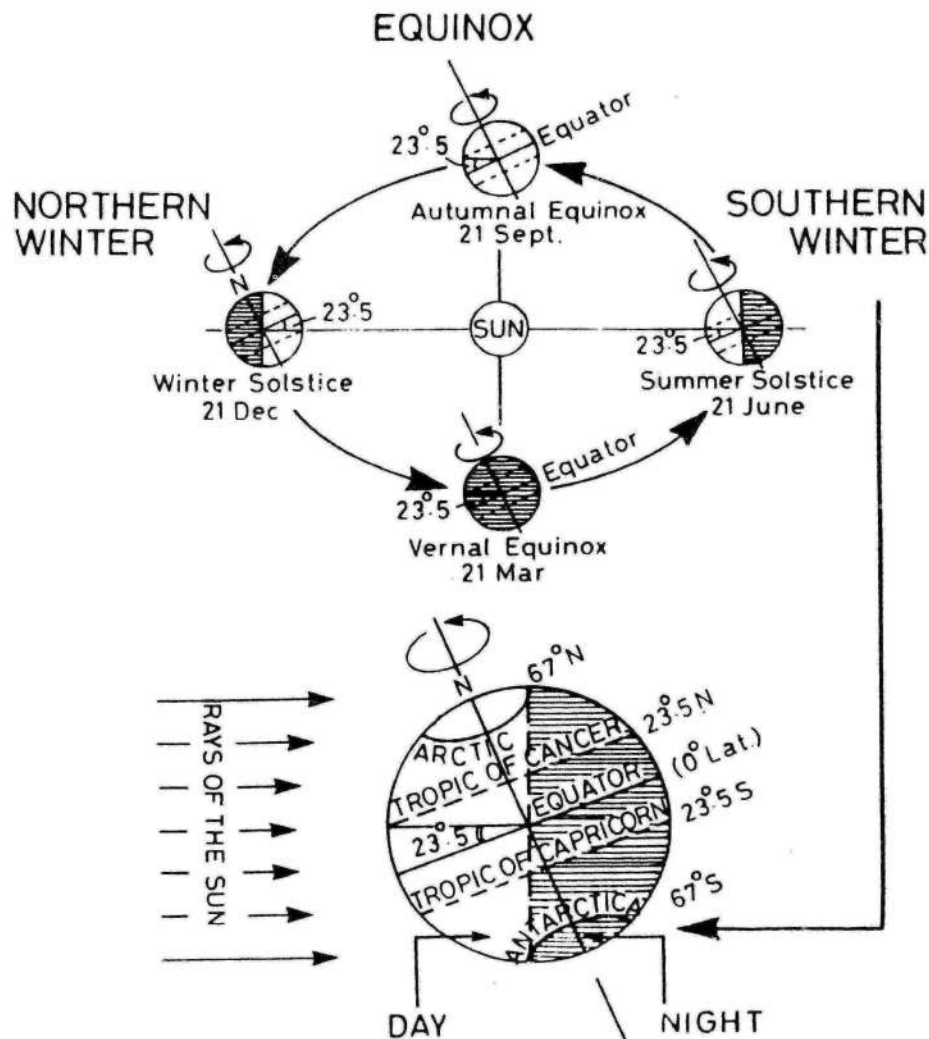


Fig. 6: Upper Block: Orbital path of planet Earth in its annual excursion around the Sun.
Lower Block: Enlarged view of Earth during the June solstice which corresponds to Antarctic Winter

between 90-120 km whereas σ_0 rises continuously with altitude, The conductivities themselves are a function of the ion/electron densities in the ionosphere and their respective mobilities; the mobilities in turn are function of the mass of the ions and electrons, and of their collision frequencies with neutral gases in the ionosphere.

Thus the current density

$$\mathbf{j} = \sigma \cdot \mathbf{E}$$

where σ – electrical conductivity in the ionosphere

$\mathbf{E}$ – electric field in the ionosphere

The detailed expressions for the Hall, Pedersen, and direct conductivities are given by Rishbeth and Garriott (1969). At high latitudes e.g. MAI, free electrons and ions are created by two sources, namely 1) solar UV and X radiation and 2) charged energetic particles precipitating into the ionosphere from the outer magnetosphere. Most of the electric current in the auroral ionosphere flows in the E region at altitudes of 100-110 km. The electrons/ions in this region are produced by X-rays in the range 8-140 Å and UV in the band 796 - 1027 Å (Allen 1965). The relative contributions of these two agents are not well understood, and they are known to vary with the solar activity cycle.

The electrons/ions produced in the ionosphere by charged particles have an altitude profile which in shape resembles that due to solar UV and X rays. Electrons of 1-10 keV cause ionisation at the E region altitudes i.e. 100-120 km altitude while protons of 10 to a few hundred keV affect this altitude region. BAILEY (1959) has worked out in detail the approximate heights of penetration in the atmosphere for vertically incident electrons, protons and helium ions.

Magnetic Quietness of Days selected in This Work

In the current work the assumption made is that the seasonal variations in the geomagnetic components, and hence in the atmospheric currents originate in seasonal variations in the solar UV radiation. This is because the days selected for study are magnetically quiet days. One must remember that the days selected are quiet by Kp standards, and Kp is a mid-latitude index of magnetic activity. The question is whether the days are quiet from the point of view of Auroral Electrojet activity. To understand this, the 3 hourly Kp values and the daily sum ΣKp for the days selected in this analysis are compared with the ΣKp indices of magnetic activity published by the Australian Antarctic station Mawson (geographic coordinates 67°36'S, 62°53'E; geomagnetic coordinates 73.2°S, 106.3°E). These are shown in Table 1.

Table 1 shows that ΣK_p at mid-latitudes for any particular day can differ considerably from K at auroral latitudes for the same day. This is illustrated by 20 Nov and 26 Dec. 1992 where K values far exceed ΣK_p values and suggests that the auroral oval is quiet disturbed magnetically, unlike the mid-latitudes. The same is true for the months June, July, August, September and October 1992. The discrepancy between using K_p and K as indices of magnetic activity is shown in Fig. 7. The figure clearly shows very active conditions in the auroral region on 20 Nov. and 26 Dec. 1992, whereas it is Quiet at mid-latitudes. This can happen during periods of substorm activity, a feature which is mainly

**Table 1: Comparison of mid latitude K_p values and auroral K values
(Mawson 67°36'S, 62°53'E)**

Quiet days of 1992 - three hourly intervals										
Day/Month	Int1	Int2	Int3	Int4	Int5	Int6	Int7	Int8	Sum	
14 Mar 92	1o	1o	2-	1o	10	1-	1-	1+	8+	K_p
	No data for Mawson									K
11 Apr 92	2-	1o	1-	0+	1-	1-	0	1-	6o	K_p
	3	3	2	0	0	0	0	0	8	K
17 May 92	1-	1-	1-	1-	0+	1-	1+	1+	6+	K_p
	1	0	1	1	0	0	1	4	8	K
2 June 92	1o	1-	1+	1+	1-	1+	1+	2+	10o	K_p
	4	2	2	2	2	2	3	5	22	K
26 Jul 92	1+	1o	1-	1+	1o	1o	1o	1-	8o	K_p
	1	1	2	1	1	2	2	3	13	K
17 Aug 92	2-	2-	0+	0+	1o	1-	1o	1+	8o	K_p
	3	3	1	0	0	0	2	5	14	K
24 Sep 92	0+	0+	0+	1-	0+	1-	1+	2+	6+	K_p
	No Data for Mawson									K
24 Oct 92	1o	1o	0	0	0	1-	0o	2-	5+	K_p
	3	2	2	0	0	1	1	4	13	K
20 Nov 92	1-	0+	1-	1o	1-	0+	2-	2+	8-	K_p
	3	3	2	2	2	2	5	4	23	K
26 Dec 92	1+	1-	1+	1+	0	1o	1o	1+	9+	K_p
	4	4	3	2	3	2	3	4	25	K

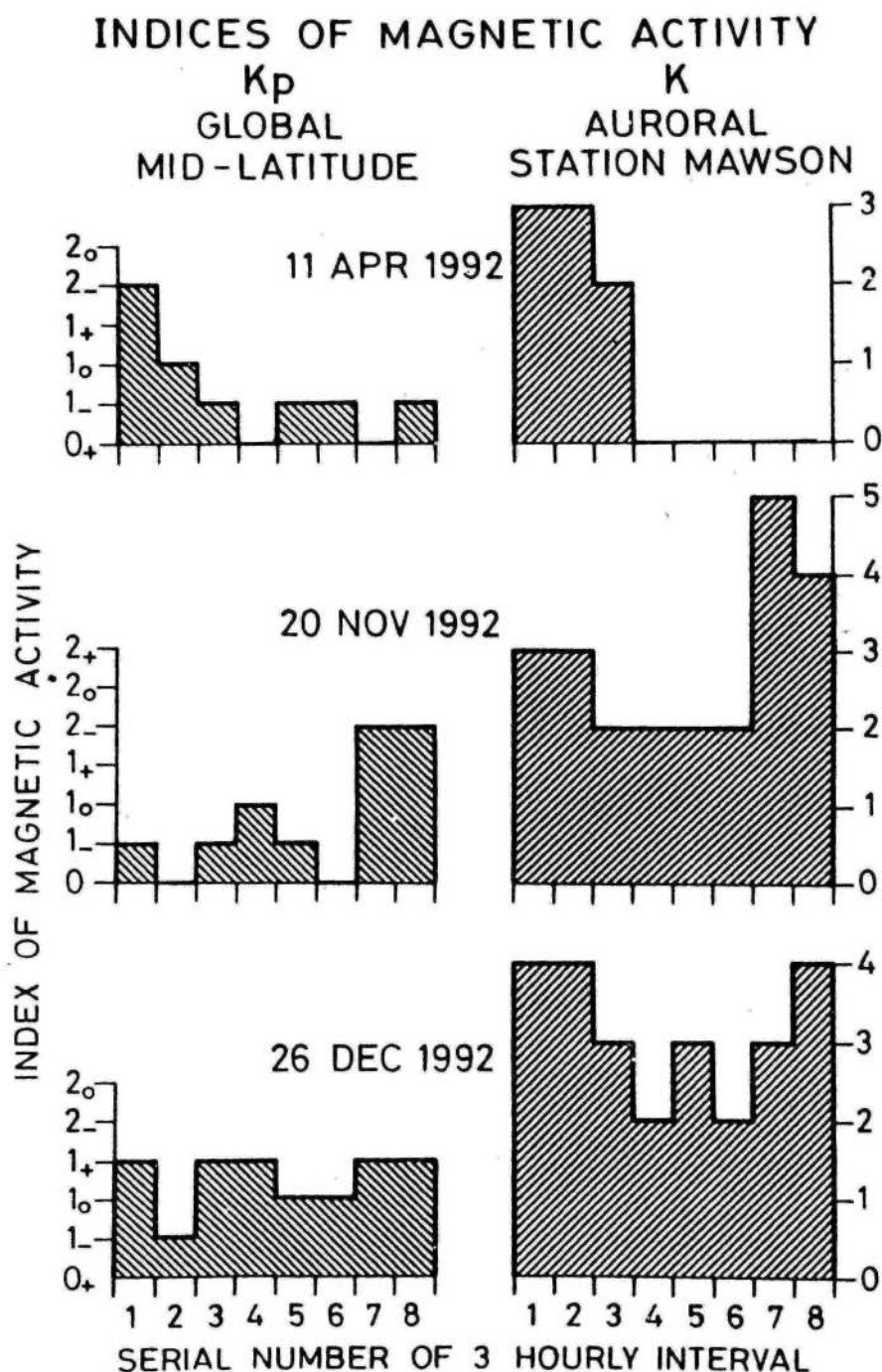


Fig. 7: A day which is deemed magnetically Quiet by the Kp index of activity need not necessarily be Quiet at auroral latitudes. This is seen by comparing the global mid-latitude index Kp with the K index of activity published by the Australian Antarctic station (Mawson)

confined to the auroral zone. Thus unusual features such as (i) the pronounced pre-midnight positive in X (Fig. 2) and X (Fig. 5) on 20 Nov. 1992, (ii) the large pre-midnight changes in Y (Fig. 4) and Z (Fig. 5) on 17 Aug 1992, could be due to enhanced auroral activity on these days which are quiet by Kp standards.

Field-aligned currents characterise the nightside of the auroral region at such times, and MAI could be experiencing the effects of the FAC on these days.

This discussion qualitatively evaluates the current systems responsible for the seasonal and other changes in the X, Y and Z geomagnetic components observed at Maitri. A more satisfying approach would be to quantitatively estimate the seasonal changes caused in ionospheric conductivity and currents by various relevant physical parameters; steps are being taken in this direction.

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